

Properties Of Solids And Liquids

JEE Main 2020 Chapterwise

Questions with Answer Keys

Physics

Q1 JEE Main 2020 - 4 September (Evening)

A cube of metal is subjected to a hydrostatic pressure of 4 GPa. The percentage change in the length of the side of the cube is close to (Given bulk modulus of metal, $B = 8 \times 10^{10} \text{ Pa}$)

- (A) 0.6
- (B) 20
- (C) 1.67
- (D) 5

Q2 JEE Main 2020 - 5 September (Evening)

In an experiment to verify Stokes law, a small spherical ball of radius r and density ρ falls under gravity through a distance h in air before entering a tank of water. If the terminal velocity of the ball inside water is same as its velocity just before entering the water surface, then the value of h is proportional to (ignore viscosity of air)

- (A) r^4
- (B) r^3
- (C) r
- (D) r^2

Q3 JEE Main 2020 - 6 September (Morning)

An object of mass m is suspended at the end of a massless wire of length L and area of cross-section, A . Young modulus of the material of the wire is Y . If the mass is pulled down slightly its frequency of oscillation along the vertical direction is

- (A) $f = \frac{1}{2\pi} \sqrt{\frac{mL}{YA}}$
- (B) $f = \frac{1}{2\pi} \sqrt{\frac{YA}{mL}}$
- (C) $f = \frac{1}{2\pi} \sqrt{\frac{YL}{mA}}$
- (D) $f = \frac{1}{2\pi} \sqrt{\frac{mA}{YL}}$

Q4 JEE Main 2020 - 7 January (Morning)

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A non-isotropic solid metal cube has coefficients of linear expansion as $5 \times 10^{-5}/^{\circ}\text{C}$ along the x -axis and $5 \times 10^{-6}/^{\circ}\text{C}$ along y -axis and z -axis. If coefficient of volume expansion of the solid is $C \times 10^{-6}/^{\circ}\text{C}$ then the value of C is

Q5 JEE Main 2020 - 9 January (Evening)

A small spherical droplet of density d is floating exactly half immerse in a liquid of density ρ and surface tension T . The radius of the droplet is (take note that the surface tension applies an upward force on the droplet):

(A) $r = \sqrt{\frac{T}{(d+\rho)g}}$

(B) $r = \sqrt{\frac{3T}{(2d-\rho)g}}$

(C) $r = \sqrt{\frac{T}{(d-\rho)g}}$

(D) $r = \sqrt{\frac{2T}{3(d+\rho)g}}$

Q6 JEE Main 2020 - 9 January (Evening)

Two steel wires having same length are suspended from a ceiling under the same load. If the ratio of their energy stored per unit volume is $1 : 4$, the ratio of their diameters is :

(A) $1 : 2$

(B) $1 : \sqrt{2}$

(C) $2 : 1$

(D) $\sqrt{2} : 1$

Answer Key

Q1 (C)

Q2 (A)

Q3 (B)

Q4 (60)

Q5 (B)

Q6 (D)

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Hints & Solutions

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Physics

Q1

$$\frac{\Delta v}{v} = \frac{\Delta p}{B} = \frac{1}{20}$$
$$\frac{\Delta I}{I} = \frac{1}{3} \frac{\Delta p}{B} = \frac{1}{60}$$
$$\frac{\Delta I}{I} \times 100 = \frac{5}{3} = 1.67\%$$

Q2

$$v_T \propto r^2$$

$$\sqrt{2gh} \propto r^2$$

$$\Rightarrow h \propto r^4$$

Q3

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$k = \frac{YA}{L}$$

$$f = \left(\frac{1}{2\pi}\right) \sqrt{\frac{YA}{mL}}$$

Q4

$$V = 2\alpha_2 + \alpha_1$$
$$= 10 \times 10^{-6} + 5 \times 10^{-5}$$
$$= 60 \times 10^{-6} / ^\circ C$$

Q5

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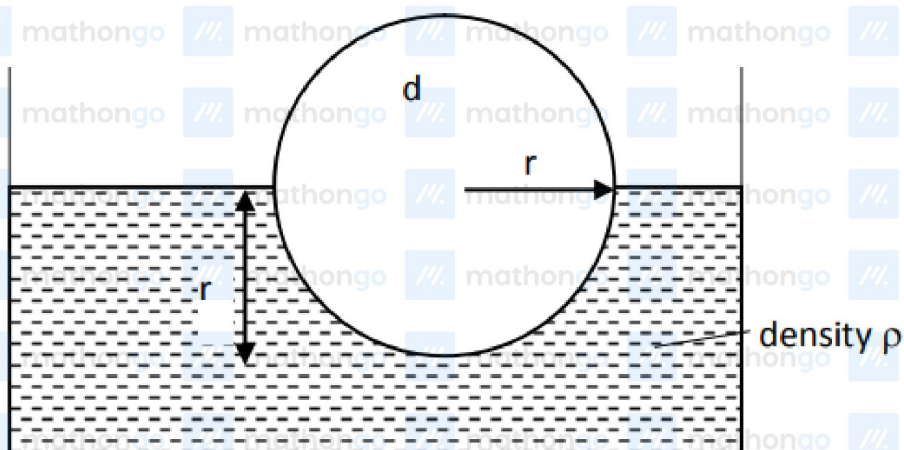
Physics

$$dVg = \rho \left(\frac{V}{2} \right) g + T(2\pi r)$$

$$\Rightarrow d \cdot \frac{4}{3}\pi r^3 g = \rho \cdot \frac{2}{3}\pi r^3 g + 2\pi r T$$

$$\Rightarrow \frac{2}{3}r^2 g (2d - \rho) = 2T$$

$$\Rightarrow r = \sqrt{\frac{3T}{(2d - \rho)g}}$$



Q6

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Physics

$$\frac{du}{dv} = \frac{1}{2} \text{stress} \times \frac{\text{stress}}{y}$$

$$= \frac{1}{2} \frac{F^2}{A^2 y}$$

$$\frac{du}{dv} \propto \frac{1}{d^4}$$

$$\frac{\left(\frac{du}{dv}\right)_1}{\left(\frac{du}{dv}\right)_2} = \frac{d_2^4}{d_1^4} = \frac{1}{4}$$

$$\frac{d_1}{d_2} = (4)^{1/4}$$

$$\frac{d_1}{d_2} = \sqrt{2} : 1$$